

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Unit Test 2

Unit Test 2

Class: Class 8

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. Which method removes and returns the last element of a list?

- b. `pop()`
- c. `remove()`
- d. `del`
- e. `discard()`

b. What will be the output?

```
t = (1, 2, 3)
t[0] = 5
```

- a. (5, 2, 3)
- b. (1, 2, 3)
- c. `TypeError`
- d. `ValueError`

c. Which of the following is valid for a dictionary?

- d. `d = {[1]: "a", [2]: "b"}`
- e. `d = {(1): "a", (2): "b"}`
- f. `d = {1: "a", 2: "b"}`
- g. `d = {"1", "a", "2", "b"}`

d. What does `dict.keys()` return?

- e. A list of values
- f. A list of key-value pairs
- g. A view object of all keys
- h. A tuple of keys

e. List comprehension to create a list of squares of numbers from 1 to 5:

- f. `[x**2 for x in range(1,6)]`
- g. `[x^2 for x in range(1,5)]`
- h. `[square(x) for x in range(1,6)]`
- i. `[x*2 for x in range(1,6)]`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write the output of:

```
nums = [10, 20, 30, 40, 50]
del nums[1:3]
print(nums)
```

Explain what happened.

b. Write a Python program to merge two dictionaries d1 = {"a": 1, "b": 2} and d2 = {"c": 3, "d": 4} and print the result.

c. What will be the output and why?

```
lst = [[1,2],[3,4],[5,6]]
for row in lst:
    print(row[1], end=" ")
```

d. Differentiate between `list.remove(x)` and `list.pop(index)`.

e. Write a program that takes a string and counts the frequency of each character using a dictionary. Display the result sorted by frequency.